



Computationally efficient system level tube-MPC for uncertain systems[☆]

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ABSTRACT

Tube-based model predictive control (MPC) is one of the principal robust control techniques for constrained linear systems affected by additive disturbances. While tube-based methods with online-computed tubes have been successfully applied to systems with additive disturbances, their application to systems affected by additional model uncertainties is challenging. This paper proposes a tube-based MPC method – named filter-based system level tube-MPC (SLTMP) – which overapproximates both types of uncertainties with an online optimized disturbance set, while simultaneously computing the tube controller online. For the first time, we provide rigorous closed-loop guarantees for receding horizon control of such a MPC method. These guarantees are obtained by virtue of a new terminal controller design and an online optimized terminal set. To reduce the computational complexity of the proposed method, we additionally introduce an asynchronous computation scheme that separates the optimization of the tube controller and the nominal trajectory. Finally, we provide a comprehensive numerical evaluation of the proposed methods to demonstrate their effectiveness.

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1. Introduction

Robust model predictive control (MPC) is one of the main robust control techniques for constrained linear systems affected by bounded uncertainties. In the presence of model uncertainty, formulating a robust MPC problem is challenging, since the predicted trajectories are affected by the uncertain model and the computed control inputs, coupling the model uncertainty and the optimization variables. Early approaches to tackle this problem were based on LMI methods (Kothare, Balakrishnan, & Morari, 1996), but more recently tube-based MPC methods have become popular for these types of uncertain systems. Tube-based MPC is based on separating the system behavior into nominal and error dynamics, which describe the system dynamics neglecting uncertainties and the deviations from the nominal dynamics due to uncertainties, respectively. Since the uncertainties are assumed to be bounded, the error dynamics can be bounded in so-called tubes, which are computed based on the uncertainty bounds and a tube controller (Rawlings, Mayne, & Diehl, 2019). However, the challenge of disentangling the model uncertainty and the optimization variables remains, which has

been addressed by mainly two different strategies in the literature. One strategy for managing both additive disturbances and model uncertainties with tube-based MPC is lumping both uncertainties into a single state- and input-dependent additive term, which is subsequently bounded offline, resulting in a system with only additive uncertainty. This approach enables the use of various well-established tube-based MPC methods. Standard tube-MPC computes the tubes offline and optimizes the nominal trajectory subject to tightened constraints online, see e.g. Chisci, Rossiter, and Zappa (2001), Mayne, Seron, and Raković (2005), Zanon and Gros (2021). To reduce conservativeness, homothetic tube-MPC (Raković, Kouvaritakis, Findeisen, & Cannon, 2012) and elastic tube-MPC (Raković, Levine, & Açikmese, 2016) enable online optimization of the tubes by fixing the tube shape, but allowing dilation and translation of the tubes online. As shown in Sieber, Zanelli, Bennani, and Zeilinger (2022), online optimization of the tube controller enables tube flexibility beyond dilation and translation. The online optimization is facilitated by parameterizing the tube controller as a disturbance feedback policy (Goulart, Kerrigan, & Maciejowski, 2006; Löfberg, 2003) or via the system level parameterization (SLP) (Sieber, Bennani, & Zeilinger, 2022), which is part of the system level synthesis framework (Anderson, Doyle, Low, & Matni, 2019). Beyond tube-based MPC, the lumped uncertainty approach is also used in classical disturbance feedback control (Bujarbaruah, Rosolia, Stürz, & Borrelli, 2021; Bujarbaruah, Rosolia, Stürz, Zhang, & Borrelli, 2022), which tightens the constraints with offline computed

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bounds as a function of the online computed control input. However, the lumped uncertainty approach is typically conservative, since it relies on offline computed global uncertainty bounds.

Another strategy treats the uncertainties separately and explicitly designs the tubes and the tube controller for the individual uncertainty descriptions. Most approaches following such a strategy build on either the homothetic tube-MPC formulation (Köhler, Andina, Soloperto, Müller, & Allgöwer, 2019; Langson, Chrysoschoos, Raković, & Mayne, 2004; Lorenzen, Cannon, & Allgöwer, 2019) or the elastic tube-MPC formulation (Fleming, Kouvaritakis, & Cannon, 2015; Lu & Cannon, 2019). Additionally, the methods proposed in Köhler et al. (2019), Lorenzen et al. (2019), Lu and Cannon (2019) incorporate online model updates based on set-membership estimation (Milanese & Viciño, 1991) to reduce conservativeness. Other tube-based methods use configuration-constrained tubes (Villanueva, Müller, & Houska, 2024), which do not explicitly parameterize a tube controller but compute a control input for every vertex of the tube, or use ellipsoidal tubes (Parsi, Iannelli, & Smith, 2022), which are computed via a LFT-based formulation of the uncertainties. More recently, Chen, Preciado, Morari, and Matni (2024) proposed a tube-based MPC method based on the SLP that simultaneously overapproximates both uncertainties and computes a tube controller online, which we denote as *filter-based system level tube-MPC* (SLTMPC). This method offers highly flexible tubes and requires minimal offline design; however, it provides closed-loop guarantees only for a shrinking horizon formulation and, like all SLP-based MPC approaches, is computationally expensive. The computational issue has been addressed on an algorithmic level by structure-exploiting solvers (Leeman et al., 2024), and on a system theoretic level by asynchronous computation schemes (Sieber, Zanelli, Leeman, Bennani, & Zeilinger, 2023), which split the tube and nominal trajectory computations into separate processes.

Contributions: The main theoretical contribution of this paper is three-fold: (i) we extend the filter-based SLTMPC method proposed in Chen et al. (2024) to general polytopic disturbance sets, which results in higher flexibility in the design and provides a more intuitive explanation of the method; (ii) we propose a new terminal controller design and an online optimized terminal set that guarantee recursive feasibility in receding horizon for filter-based SLTMPC; and (iii) we propose a new asynchronous computation scheme that reduces the computational complexity of filter-based SLTMPC methods by splitting the tube and nominal trajectory optimizations into different processes. Additionally, we provide a comprehensive numerical evaluation of the proposed methods on a double integrator and a vertical take-off and landing (VTOL) vehicle.

The paper is organized as follows: Section 2 introduces the notation, the problem formulation, and some basic definitions. Section 3 introduces filter-based SLTMPC generalized from Chen et al. (2024), before Section 4 introduces the new terminal ingredients and proves recursive feasibility. Section 5 details the asynchronous computation scheme and Section 6 showcases the proposed filter-based SLTMPC variants on two numerical examples. Finally, Section 7 concludes the paper.

An extended version of this paper (Sieber, Didier, & Zeilinger, 2024) including the full proofs and all implementation details is available online. Additionally, the code for the numerical examples is available at git.sieber.io/sltmpc.

2. Preliminaries

2.1. Notation & definitions

We indicate column vectors, e.g. a , and matrices, e.g. A , with lowercase and uppercase letters, respectively. We denote stacked

column vectors, e.g. $\mathbf{a}$, and block matrices, e.g., $\mathbf{A}$, with bold letters and index their (block) elements with subscripts, e.g., a_i , $A_{i,j}$. We use $\text{diag}(\mathbf{A})$ to denote the block-diagonal matrix, whose diagonal consists of i -times the A matrix. We define the operator $\text{shift}(\mathbf{A}, \mathbf{B})$ for a block-lower-triangular matrix $\mathbf{A}$ with $N \times N$ block elements and a block row matrix $\mathbf{B}$ with N block elements, i.e. $\mathbf{B} = [B_0 \dots B_{N-1}]$, as

$$\text{shift}(\mathbf{A}, \mathbf{B}) = \begin{bmatrix} A_{1,1} & & & \\ \vdots & \ddots & & \\ A_{N-1,1} & \dots & A_{N-1,N-1} & \\ B_0 & \dots & \dots & B_{N-1} \end{bmatrix},$$

which removes the first row and column of $\mathbf{A}$ and inserts $\mathbf{B}$ in the bottom block row. We use $\mathbb{I}_n$ to denote the identity matrix of size $n \times n$. We distinguish between the states of a dynamical system $x(k)$ and the states predicted by an MPC algorithm x_i . We use $\oplus$ and $\ominus$ to denote Minkowski set addition and Pontryagin set subtraction, respectively, which are both formally defined in Rawlings et al. (2019, Definition 3.10). For a sequence of sets $\mathcal{A}_0, \dots, \mathcal{A}_j$, we use the convention that $\bigoplus_{i=0}^{-1} \mathcal{A}_i = \{0\}$, which is the set containing only the zero vector. For $\mathcal{K}$ -functions and uniform continuity we use the standard definitions in Rawlings et al. (2019).

2.2. Problem formulation

We consider linear time-invariant (LTI) dynamical systems with parametric and additive uncertainties, i.e.,

$$x(k+1) = (A + \Delta_A)x(k) + (B + \Delta_B)u(k) + w(k), \quad (1)$$

with $A \in \mathbb{R}^{n \times n}$, $B \in \mathbb{R}^{n \times m}$, $(\Delta_A, \Delta_B) \in \mathcal{D}$, and $w(k) \in \mathcal{W} \subset \mathbb{R}^n$, where $\mathcal{W}$ and $\mathcal{D}$ are the convex polytopes

$$\mathcal{W} = \{w \in \mathbb{R}^n \mid H_w w \leq h_w\}, \quad (2a)$$

$$\mathcal{D} = \text{co}\{(\Delta_A^d, \Delta_B^d)\}, \quad d = 1, \dots, n_D, \quad (2b)$$

where $H_w \in \mathbb{R}^{n_w \times n}$, $h_w \in \mathbb{R}^{n_w}$, and n_D is the number of generators for Δ_A and Δ_B . The set $\mathcal{D}$ is thus defined by the vertices $\mathcal{D}^d = (\Delta_A^d, \Delta_B^d)$, $d = 1, \dots, n_D$. The system is subject to compact polytopic state and input constraints

$$\mathcal{X} = \{x \in \mathbb{R}^n \mid H_x x \leq h_x\}, \quad \mathcal{U} = \{u \in \mathbb{R}^m \mid H_u u \leq h_u\}, \quad (3)$$

where $H_x \in \mathbb{R}^{n_x \times n}$, $h_x \in \mathbb{R}^{n_x}$, $H_u \in \mathbb{R}^{n_u \times m}$, $h_u \in \mathbb{R}^{n_u}$, with both sets containing the origin in their interior. To safely control system (1), we consider the following robust constraint satisfaction problem (CSP) over the task horizon $\tilde{N}$.

Problem 1 (Robust CSP). Find a sequence of inputs u_i , $i = 0, \dots, \tilde{N} - 1$, such that the following constraints are satisfied for all $i = 0, \dots, \tilde{N} - 1$:

$$x_0 = x(k), \quad (4a)$$

$$x_{i+1} = (A + \Delta_A)x_i + (B + \Delta_B)u_i + w_i, \quad (4b)$$

$$x_i \in \mathcal{X}, \quad \forall (\Delta_A, \Delta_B) \in \mathcal{D}, \quad \forall w_i \in \mathcal{W}, \quad (4c)$$

$$u_i \in \mathcal{U}, \quad (4d)$$

$$x_{\tilde{N}} \in \mathcal{S}_f, \quad (4e)$$

where x_i , u_i , and w_i denote the predicted state, input, and disturbance, respectively, and $\mathcal{S}_f \subseteq \mathcal{X}$ is a robust positively invariant (RPI) set according to Definition 1.

Definition 1 (RPI Set for (1)). The set $\mathcal{S}_f \subseteq \mathcal{X}$ is a robust positively invariant (RPI) set for system (1) with control law $u = K_f x \in \mathcal{U}$ for all $x \in \mathcal{S}_f$, if $x \in \mathcal{S}_f \implies x^+ \in \mathcal{S}_f$ for all $w \in \mathcal{W}$, $(\Delta_A, \Delta_B) \in \mathcal{D}$.

In this paper, we solve the CSP using a tube-based MPC formulation (Rawlings et al., 2019; Sieber, Bennani, & Zeilinger, 2022) that tightens the constraints with so-called tubes for a shorter horizon $N < \bar{N}$ and apply the MPC in shrinking horizon or receding horizon. In the following section, we detail how to derive such a formulation by generalizing the tube-based MPC formulation proposed in Chen et al. (2024).

3. Filter-based system level tube-MPC

The filter-based SLTMPC proposed in Chen et al. (2024) relies on designing an auxiliary disturbance set $\bar{\mathcal{W}}$, before transforming it online to overapproximate the combined effect of additive and parametric uncertainties. This enables online adaptation of the tube controller and results in a nonconservative tube-based MPC method. We extend this method to general polytopic disturbance sets $\bar{\mathcal{W}} = \{\bar{w} \in \mathbb{R}^n \mid H_{\bar{w}} \bar{w} \leq h_{\bar{w}}\}$, with $H_{\bar{w}} \in \mathbb{R}^{n_{\bar{w}} \times n}$, $h_{\bar{w}} \in \mathbb{R}^{n_{\bar{w}}}$, instead of norm-bounded disturbance sets $\bar{\mathcal{W}} = \{\bar{w} \in \mathbb{R}^n \mid \|\bar{w}\|_{\infty} \leq 1\}$, which extends the scope of the method and simplifies its exposition. We first rewrite the prediction dynamics (4b) by collecting all uncertainties in a separate term denoted as η_i :

$$x_{i+1} = Ax_i + Bu_i + \eta_i, \quad (5)$$

$$\eta_i = \Delta_A x_i + \Delta_B u_i + w_i. \quad (6)$$

Following the idea presented in Chen et al. (2024), we then overapproximate the combined uncertainty η_i for $i = 0, \dots, N-1$ with online-computed *disturbance tubes*, i.e.,

$$\eta_i \in \{p_i\} \oplus \mathcal{F}_i(\Sigma) := \{p_i\} \oplus \bigoplus_{j=0}^i \Sigma_{i+1,j} \bar{\mathcal{W}}, \quad (7)$$

where $p_i \in \mathbb{R}^n$, $\Sigma_{i+1,j} \in \mathbb{R}^{n \times n}$ are online optimization variables denoting the nominal disturbance and the disturbance filter, respectively. The compact polytopic disturbance set $\bar{\mathcal{W}} \subset \mathbb{R}^n$, which contains the origin in its interior, is designed offline. Note that these disturbance tubes (7) are a generalization of homothetic tubes (Raković et al., 2012). This can be seen by setting $\Sigma_{i+1,i} = \alpha_i \cdot \mathbb{I}_n$ and $\Sigma_{i+1,j} = 0$ for $j = 0, \dots, i-1$, which simplifies containment condition (7) to $\eta_i \in \{p_i\} \oplus \alpha_i \bar{\mathcal{W}}$, where α_i are the dilation factors and p_i are the tube centers in Raković et al. (2012). If we enforce (7) for all $w \in \mathcal{W}$, $(\Delta_A, \Delta_B) \in \mathcal{D}$ during online optimization, we can guarantee that for optimized p_i , $\Sigma_{i+1,j}$, there always exists a sequence of $\bar{w}_j \in \bar{\mathcal{W}}$, $j = 0, \dots, i$ such that

$$\eta_i = p_i + \sum_{j=0}^i \Sigma_{i+1,j} \bar{w}_j, \quad i = 0, \dots, N-1.$$

Therefore, we can equivalently write dynamics (5) as

$$x_{i+1} = Ax_i + Bu_i + p_i + \sum_{j=0}^i \Sigma_{i+1,j} \bar{w}_j, \quad (8)$$

when imposing that (7) holds.

Remark 1. The application of disturbance tubes (7) is not limited to system (1) with uncertainty description $\mathcal{W}$, $\mathcal{D}$, but can be extended to other uncertainty descriptions, e.g., the one used in Köhler et al. (2019), as long as the combined uncertainty η_i is a linear function of x_i and u_i .

In the following, we use auxiliary dynamics (8) and formulate the filter-based SLTMPC problem by splitting (8) into nominal and error dynamics. We use z_i and e_i as the nominal and the error state, respectively, such that $x_i = z_i + e_i$. Together with the tube-based control policy $u_i = v_i + v_i$, where v_i and v_i are the nominal

and error inputs, we obtain

$$x_{i+1} = A(z_i + e_i) + B(v_i + v_i) + p_i + \sum_{j=0}^i \Sigma_{i+1,j} \bar{w}_j,$$

and thus – as is standard in tube-based MPC (Rawlings et al., 2019) – separation into nominal and error dynamics, i.e.,

$$z_{i+1} = Az_i + Bv_i + p_i, \quad (9)$$

$$e_{i+1} = Ae_i + Bv_i + \sum_{j=0}^i \Sigma_{i+1,j} \bar{w}_j. \quad (10)$$

Since we consider MPC with horizon length N , we define the trajectories $\mathbf{z} = [z_0^\top, \dots, z_N^\top]^\top$, $\mathbf{v} = [v_0^\top, \dots, v_N^\top]^\top$, $\mathbf{p} = [p_0^\top, \dots, p_{N-1}^\top]^\top$, $\bar{\mathbf{w}} = [\bar{w}_0^\top, \dots, \bar{w}_{N-1}^\top]^\top$, $\mathbf{e} = [e_1^\top, \dots, e_N^\top]^\top$, and $\mathbf{v} = [v_1^\top, \dots, v_N^\top]^\top$ as the nominal state, nominal input, nominal disturbance, auxiliary disturbance, error state, and error input trajectories, respectively. Using these trajectories, the nominal dynamics are then compactly written as

$$\mathbf{z} = \mathbf{Z}\mathbf{A}_{N+1}\mathbf{z} + \mathbf{Z}\mathbf{B}_{N+1}\mathbf{v} + \begin{bmatrix} z_0 \\ \mathbf{p} \end{bmatrix}, \quad (11)$$

and the error dynamics as

$$\mathbf{e} = \mathbf{Z}\mathbf{A}_N\mathbf{e} + \mathbf{Z}\mathbf{B}_N\mathbf{v} + \Sigma\bar{\mathbf{w}}, \quad (12)$$

where $\mathbf{Z}$ is a lower shift matrix, i.e., a matrix with ones on its first subdiagonal and zeros everywhere else, $\mathbf{A}_N = \text{diag}_N(A)$, $\mathbf{B}_N = \text{diag}_N(B)$, and $\Sigma \in \mathbb{R}^{Nn \times Nn}$ is a block-lower-triangular *disturbance filter*. This filter and the time-varying feedback controller $\mathbf{K} \in \mathbb{R}^{Nn \times Nn}$ in the error input $\mathbf{v} = \mathbf{K}\mathbf{e}$ are defined as

$$\Sigma = \begin{bmatrix} \Sigma_{1,0} & & & \\ \Sigma_{2,0} & \Sigma_{2,1} & & \\ \vdots & \vdots & \ddots & \\ \Sigma_{N,0} & \Sigma_{N,1} & \dots & \Sigma_{N,N-1} \end{bmatrix}, \quad \mathbf{K} = \begin{bmatrix} K_{1,1} & & & \\ K_{2,1} & K_{2,2} & & \\ \vdots & \vdots & \ddots & \\ K_{N,1} & K_{N,2} & \dots & K_{N,N} \end{bmatrix}.$$

Substituting the error input $\mathbf{v}$ into (12) yields

$$\mathbf{e} = (\mathbf{Z}\mathbf{A}_N + \mathbf{Z}\mathbf{B}_N\mathbf{K})\mathbf{e} + \Sigma\bar{\mathbf{w}}, \quad (13)$$

$$\Leftrightarrow \mathbf{e} = (\mathbb{I}_{Nn} - \mathbf{Z}\mathbf{A}_N - \mathbf{Z}\mathbf{B}_N\mathbf{K})^{-1} \Sigma\bar{\mathbf{w}}.$$

Note that the inverse in (13) always exists, since it admits a finite Neumann series due to the lower shift matrix $\mathbf{Z}$. As in Chen et al. (2024), we use the filter-based system level parameterization (SLP) to reparameterize (13) and the associated error input $\mathbf{v}$ as

$$\mathbf{e} = (\mathbb{I}_{Nn} - \mathbf{Z}\mathbf{A}_N - \mathbf{Z}\mathbf{B}_N\mathbf{K})^{-1} \Sigma\bar{\mathbf{w}} = \Phi^e \bar{\mathbf{w}}, \quad (14a)$$

$$\mathbf{v} = \mathbf{K}(\mathbb{I}_{Nn} - \mathbf{Z}\mathbf{A}_N - \mathbf{Z}\mathbf{B}_N\mathbf{K})^{-1} \Sigma\bar{\mathbf{w}} = \Phi^v \bar{\mathbf{w}}, \quad (14b)$$

where $\Phi^e \in \mathbb{R}^{Nn \times Nn}$, $\Phi^v \in \mathbb{R}^{Nm \times Nn}$ are the *error system responses*, which admit the following block-lower-triangular structure

$$\Phi^e = \begin{bmatrix} \Phi_{1,0}^e & & & \\ \Phi_{2,0}^e & \Phi_{2,1}^e & & \\ \vdots & \vdots & \ddots & \\ \Phi_{N,0}^e & \Phi_{N,1}^e & \dots & \Phi_{N,N-1}^e \end{bmatrix}, \quad \Phi^v = \begin{bmatrix} \Phi_{1,0}^v & & & \\ \Phi_{2,0}^v & \Phi_{2,1}^v & & \\ \vdots & \vdots & \ddots & \\ \Phi_{N,0}^v & \Phi_{N,1}^v & \dots & \Phi_{N,N-1}^v \end{bmatrix}.$$

Using Chen et al. (2024, Corollary 1) we guarantee that the error system responses Φ^e , Φ^v parameterize all error trajectories $\mathbf{e}$, $\mathbf{v}$ that are realized by tube controller $\mathbf{K}$ and disturbance filter Σ . Using the filter-based SLP (14), the error dynamics (12) are equivalently rewritten as

$$\begin{bmatrix} \mathbb{I}_{Nn} - \mathbf{Z}\mathbf{A}_N & -\mathbf{Z}\mathbf{B}_N \end{bmatrix} \begin{bmatrix} \Phi^e \\ \Phi^v \end{bmatrix} = \Sigma. \quad (15)$$

Thereby, the error dynamics are completely defined by the closed-loop behavior, which allows optimization over the system responses instead of optimizing tube controller. Note that the filter-based SLP (14) can also be written in elementwise form, i.e., for $i = 1, \dots, N$

$$e_i = \sum_{j=0}^{i-1} \Phi_{i,j}^e \bar{w}_j, \quad v_i = \sum_{j=0}^{i-1} \Phi_{i,j}^v \bar{w}_j, \quad (16)$$

and that (15) constrains the block diagonals of Φ^e and Σ due to the lower shift matrices $\mathbf{Z}$, i.e., it holds that $\Phi_{i+1,i}^e = \Sigma_{i+1,i}$ for all $i = 0, \dots, N-1$. In Chen et al. (2024, Section 4) it is shown how to enforce containment condition (7) for norm-bounded auxiliary disturbance sets $\bar{\mathcal{W}}$. We extend this analysis to general polytopic disturbance sets $\bar{\mathcal{W}}$ in the following. First, we split the state x_i and input u_i in (6) into their nominal and error contributions and apply the elementwise SLP (16) to the error state e_i and error input v_i , i.e.,

$$\begin{aligned} \eta_i &= \Delta_A(z_i + e_i) + \Delta_B(v_i + v_i) + w_i \\ &= \Delta_A z_i + \Delta_B v_i + \sum_{j=0}^{i-1} (\Delta_A \Phi_{i,j}^e + \Delta_B \Phi_{i,j}^v) \bar{w}_j + w_i. \end{aligned}$$

The containment condition (7) for $i = 0, \dots, N-1$ is then

$$\begin{aligned} \{\Delta_A z_i + \Delta_B v_i\} \oplus \bigoplus_{j=0}^{i-1} (\Delta_A \Phi_{i,j}^e + \Delta_B \Phi_{i,j}^v) \bar{\mathcal{W}} \oplus \mathcal{W} \\ \subseteq \{p_i\} \oplus \bigoplus_{j=0}^i \Sigma_{i+1,j} \bar{\mathcal{W}}, \quad \forall (\Delta_A, \Delta_B) \in \mathcal{D}, \end{aligned}$$

and by collecting the singletons we get

$$\begin{aligned} \{\Delta_A z_i + \Delta_B v_i - p_i\} \oplus \bigoplus_{j=0}^{i-1} (\Delta_A \Phi_{i,j}^e + \Delta_B \Phi_{i,j}^v) \bar{\mathcal{W}} \oplus \mathcal{W} \\ \subseteq \bigoplus_{j=0}^i \Sigma_{i+1,j} \bar{\mathcal{W}} = \mathcal{F}_i(\Sigma), \quad \forall (\Delta_A, \Delta_B) \in \mathcal{D}. \end{aligned} \quad (17)$$

Since the superset of above inclusion is a Minkowski sum, embedding it in an optimization problem is conservative and non-convex (Sadraddini & Tedrake, 2019). Therefore, we make use of the following lemma to convert (17) into a form that can be formulated linearly in the optimization variables.

Lemma 1. Let $\mathcal{D}^d = (\Delta_A^d, \Delta_B^d)$ denote the d^{th} vertex of $\mathcal{D}$ and define

$$\psi_i^d := \Delta_A^d z_i + \Delta_B^d v_i - p_i, \quad (18a)$$

$$\Psi_{i,j}^d := \Delta_A^d \Phi_{i,j}^e + \Delta_B^d \Phi_{i,j}^v - \Sigma_{i+1,j}. \quad (18b)$$

Then, if the following inclusion holds for $i = 0, \dots, N-1$ and $d = 1, \dots, n_D$,

$$\{\psi_i^d\} \oplus \bigoplus_{j=0}^{i-1} \Psi_{i,j}^d \bar{\mathcal{W}} \oplus \mathcal{W} \subseteq \Sigma_{i+1,i} \bar{\mathcal{W}}, \quad (19)$$

containment condition (17) is satisfied.

Proof. First, note that due to the separation $x = z + e$ and $u = v + v$, the same Δ_A^d and Δ_B^d are applied to the nominal and error variables. Therefore, the same vertex of $\mathcal{D}$ is used in (18a), (18b) and we can prove the lemma for any vertex d , implying that the proof holds for all $(\Delta_A, \Delta_B) \in \mathcal{D}$ due to convexity of $\mathcal{D}$. We start by rewriting (17) using (18) as

$$\{\psi_i^d\} \oplus \bigoplus_{j=0}^{i-1} (\Psi_{i,j}^d + \Sigma_{i+1,j}) \bar{\mathcal{W}} \oplus \mathcal{W} \subseteq \bigoplus_{j=0}^i \Sigma_{i+1,j} \bar{\mathcal{W}}, \quad (20)$$

where we used $\Psi_{i,j}^d + \Sigma_{i+1,j} = \Delta_A^d \Phi_{i,j}^e + \Delta_B^d \Phi_{i,j}^v$. Next, we note that the following inclusion holds

$$(\psi_i^d + \Sigma_{i+1,i}) \bar{\mathcal{W}} \subseteq \Psi_{i,j}^d \bar{\mathcal{W}} \oplus \Sigma_{i+1,j} \bar{\mathcal{W}}, \quad (21)$$

which is a direct consequence of Sadraddini and Tedrake (2019, Proposition 2). Finally, we show (20) holds by using (21) and commutativity of the Minkowski sum, i.e.,

$$\begin{aligned} \{\psi_i^d\} \oplus \bigoplus_{j=0}^{i-1} (\Psi_{i,j}^d + \Sigma_{i+1,j}) \bar{\mathcal{W}} \oplus \mathcal{W} \\ \stackrel{(21)}{\subseteq} \{\psi_i^d\} \oplus \bigoplus_{j=0}^{i-1} \Psi_{i,j}^d \bar{\mathcal{W}} \oplus \mathcal{W} \oplus \bigoplus_{j=0}^{i-1} \Sigma_{i+1,j} \bar{\mathcal{W}} \\ \stackrel{(19)}{\subseteq} \Sigma_{i+1,i} \bar{\mathcal{W}} \oplus \bigoplus_{j=0}^{i-1} \Sigma_{i+1,j} \bar{\mathcal{W}} \\ = \bigoplus_{j=0}^i \Sigma_{i+1,j} \bar{\mathcal{W}}. \end{aligned}$$

Hence, we have shown that enforcing (19) is sufficient for (17) to hold. $\square$

Additionally, we impose a diagonal structure on Σ , i.e., $\Sigma_{i+1,i} = \sigma_i \cdot \mathbb{I}_n$ for $i = 0, \dots, N-1$, which is similarly done in Chen et al. (2024, Section 4.1). Using the correspondence to homothetic tubes (Raković et al., 2012), we note that this diagonal structure with scalings σ_i is equivalent to the dilation factors α_i . After applying Lemma 1 and imposing the diagonal structure, the inclusion (17) for $i = 0, \dots, N-1$ and $d = 1, \dots, n_D$ reads as

$$\{\psi_i^d\} \oplus \bigoplus_{j=0}^{i-1} \Psi_{i,j}^d \bar{\mathcal{W}} \oplus \mathcal{W} \subseteq \sigma_{i+1} \bar{\mathcal{W}}. \quad (22)$$

Remark 2. Note that we can relax the structural constraint $\Sigma_{i+1,i} = \sigma_i \cdot \mathbb{I}_n$ to $\Sigma_{i+1,i} = \text{diag}(\sigma_{i,1}, \dots, \sigma_{i,n})$, if we design $\bar{\mathcal{W}}$ as a hyperrectangle, therefore recovering a similar disturbance tube parameterization as in Chen et al. (2024).

Finally, we use nominal dynamics (9), error dynamics (15), and disturbance overapproximation (22) to formulate the generalized version of the filter-based SLTPC problem (Chen et al., 2024) as

$$\min_{\substack{z, v, p, \Sigma, \\ \Phi^e, \Phi^v}} l_f(z_N) + \sum_{i=0}^{N-1} l(z_i, v_i), \quad (23a)$$

$$\text{s.t. } \forall i = 0, \dots, N-1:$$

$$z_0 = x(k), \quad (23b)$$

$$z_{i+1} = Az_i + Bv_i + p_i, \quad (23c)$$

$$\begin{bmatrix} \mathbb{I}_{Nn} - \mathbf{Z}\mathbf{A}_N & -\mathbf{Z}\mathbf{B}_N \end{bmatrix} \begin{bmatrix} \Phi^e \\ \Phi^v \end{bmatrix} = \Sigma, \quad (23d)$$

$$z_i \in \mathcal{X} \ominus \mathcal{F}_i(\Phi^e), \quad (23e)$$

$$v_i \in \mathcal{U} \ominus \mathcal{F}_i(\Phi^v), \quad (23f)$$

$$z_N \in \mathcal{S}_f \ominus \mathcal{F}_N(\Phi^e), \quad (23g)$$

$$\{\psi_i^d\} \oplus \bigoplus_{j=0}^{i-1} \Psi_{i,j}^d \bar{\mathcal{W}} \oplus \mathcal{W} \subseteq \sigma_{i+1} \bar{\mathcal{W}}, \quad d = 1, \dots, n_D, \quad (23h)$$

where $l(\cdot, \cdot)$ and $l_f(\cdot)$ are suitable stage and terminal costs, ψ_i^d and $\Psi_{i,j}^d$ are defined as in (18), $\mathcal{S}_f$ is an RPI terminal set according

to Definition 1, and $\mathcal{F}_i(\Phi^e)$, $\mathcal{F}_i(\Phi^v)$ are the state and input tubes defined as

$$\mathcal{F}_i(\Phi^e) := \bigoplus_{j=0}^{i-1} \Phi_{i,j}^e \bar{W}, \quad \mathcal{F}_i(\Phi^v) := \bigoplus_{j=0}^{i-1} \Phi_{i,j}^v \bar{W}. \quad (24)$$

In order to show recursive feasibility of (23) we would need to show that $\mathcal{S}_f \ominus \mathcal{F}_N(\Phi^e)$ is RPI itself. However, since $\mathcal{S}_f$ is RPI with respect to both $w \in \mathcal{W}$ and $(\Delta_A, \Delta_B) \in \mathcal{D}$ (Definition 1) this is difficult because $\mathcal{F}_N(\Phi^e)$ would need to be an exact reachable set of (1) for all $w \in \mathcal{W}$, $(\Delta_A, \Delta_B) \in \mathcal{D}$. This is clearly not the case, since $\mathcal{F}_N(\Phi^e)$ is computed via (17), which is an overapproximation of the combined uncertainties. Therefore, we need to restrict the SLTMPC (23) to a shrinking horizon regime similar to Bujarbaruah et al. (2022), Chen et al. (2024), which switch between solving the MPC problem with a shrinking horizon and exactly solving the robust CSP (4) for $\bar{N} = 1$ to show recursive feasibility and robust stability. However, this strategy requires implementation of a switching logic and only works well if task horizon $\bar{N}$ is finite and known in advance. In the next section, we show how to modify the terminal constraints in (23) such that the resulting MPC can be applied in receding horizon. For a recursive feasibility proof of (23) in shrinking horizon, we refer to Bujarbaruah et al. (2022, Appendix 6).

4. Recursively feasible filter-based SLTMPC

The terminal constraints in (23) are not suitable to prove recursive feasibility in receding horizon, due to $\mathcal{S}_f$ being computed for system (1) with combined uncertainty η , and $\mathcal{F}_N(\Phi^e)$ being computed for auxiliary system (8) with only additive uncertainty $\bar{w}$. Therefore, we propose a new set of terminal constraints that only rely on sets computed for auxiliary system (8). The key idea is to exploit the separation $x = z + e$ and formulate a separate terminal control law for both the nominal and error states, ensuring that both only depend on $\bar{w}$. The resulting new terminal set $\mathcal{X}_f$ is then used to constrain the terminal state $x_N \in \mathcal{X}_f$ in the proposed MPC scheme.

For this we first define an auxiliary RPI set $\mathcal{Z}_f$ for a simplified version of (8) with $p_i = 0$, $\Sigma_{i+1,i} = \mathbb{I}_n$, $\Sigma_{i+1,j} = 0$ for all i and $j = 0, \dots, i-1$, i.e.,

$$x(k+1) = Ax(k) + Bu(k) + \bar{w}(k). \quad (25)$$

Definition 2 (RPI Set for (25)). The set $\mathcal{Z}_f \subseteq \mathcal{X}$ is a robust positively invariant (RPI) set for system (25) with control law $u = K_f x \in \mathcal{U}$ for all $x \in \mathcal{Z}_f$, if $x \in \mathcal{Z}_f \implies x^+ \in \mathcal{Z}_f$ for all $\bar{w}(k) \in \bar{\mathcal{W}}$.

We then use the control law K_f of RPI set $\mathcal{Z}_f$ to construct the terminal control law as

$$\kappa_f(x) = \kappa_f^z(z) + \kappa_f^e(e) = K_f z + v = K_f z + \sum_{j=0}^N \Phi_{N,j}^v \bar{w}_{N-j}, \quad (26)$$

where $\Phi_{N,j}^v$, i.e. the last block row of Φ^v , can be freely optimized since it is not constrained by (15). Before constructing the RPI set corresponding to control law (26), we first overapproximate the terminal uncertainty η_N with a terminal disturbance tube similar to (7),¹ i.e.,

$$\eta_N \in \mathcal{F}_N(\Xi) := \bigoplus_{j=0}^{N-1} \Xi_j \bar{W} \oplus \sigma_N \bar{W}, \quad (27)$$

¹ Note that $i = N$ is not included in (7), hence we need to define this case separately.

where $\Xi = [\Xi_0 \dots \Xi_{N-1}]$ are additional disturbance filter parameters. Similar to Section 3, there always exists a sequence of $\bar{w}_j \in \bar{\mathcal{W}}$, $j = 0, \dots, N$, such that $\eta_N = \sum_{j=0}^{N-1} \Xi_j \bar{w}_j + \sigma_N \bar{w}_N$. With this, we define RPI set $\mathcal{X}_f$ for control law (26) in the following lemma.

Lemma 2. The set $\mathcal{X}_f := \alpha \mathcal{Z}_f \oplus \mathcal{F}_N(\Phi^e)$, with $\alpha \geq 0$, is an RPI set for system $x_{N+1} = Ax_N + B\kappa_f(x_N) + \eta_N$ with control law $\kappa_f(\cdot)$ defined in (26) and $\eta_N \in \mathcal{F}_N(\Xi)$, if the following two conditions hold:

- (a) $\alpha(A + BK_f)\mathcal{Z}_f \oplus \Gamma\bar{\mathcal{W}} \subseteq \alpha\mathcal{Z}_f$ with $\mathcal{Z}_f$ RPI according to Definition 2 and $\Gamma = A\Phi_{N,0}^e + B\Phi_{N,0}^v + \Xi_0$,
- (b) $\Phi_{N,j-1}^e = A\Phi_{N,j}^e + B\Phi_{N,j}^v + \Xi_j$, $\forall j = 1, \dots, N-1$.

It holds that if $x_N \in \mathcal{X}_f \implies x_{N+1} \in \mathcal{X}_f \forall \eta_N \in \mathcal{F}_N(\Xi)$.

Proof. We start by analyzing the dynamics under control law (26), i.e.,

$$x_{N+1} = (A + BK_f)z_N + Ae_N + Bv_N + \eta_N.$$

Next, we insert the definitions of e_N , v_N , and η_N to get

$$\begin{aligned} Ae_N + Bv_N + \eta_N &= \sum_{j=0}^{N-1} (A\Phi_{N,j}^e + B\Phi_{N,j}^v + \Xi_j) \bar{w}_j + \sigma_N \bar{w}_N \\ &= (A\Phi_{N,0}^e + B\Phi_{N,0}^v + \Xi_0) \bar{w}_0 \\ &\quad + \sum_{j=1}^{N-1} (A\Phi_{N,j}^e + B\Phi_{N,j}^v + \Xi_j) \bar{w}_j + \sigma_N \bar{w}_N. \end{aligned}$$

Using $\Phi_{N,N-1}^e = \Sigma_{N,N-1} = \sigma_N \cdot \mathbb{I}_n$ due to (15), $\Gamma = A\Phi_{N,0}^e + B\Phi_{N,0}^v + \Xi_0$ from (a), and (b) we obtain

$$\begin{aligned} Ae_N + Bv_N + \eta_N &= \Gamma \bar{w}_0 + \sum_{j=1}^{N-1} \Phi_{N,j-1}^e \bar{w}_j + \Phi_{N,N-1}^e \bar{w}_N \\ &= \Gamma \bar{w}_0 + \sum_{j=0}^{N-1} \Phi_{N,j}^e \bar{w}_{j+1}. \end{aligned}$$

Therefore, the dynamics under control law (26) are

$$x_{N+1} = \underbrace{(A + BK_f)z_N + \Gamma \bar{w}_0}_{:=z_{N+1}} + \underbrace{\sum_{j=0}^{N-1} \Phi_{N,j}^e \bar{w}_{j+1}}_{:=e_{N+1}}.$$

Now, if $x_N = z_N + e_N \in \alpha \mathcal{Z}_f \oplus \mathcal{F}_N(\Phi^e) = \mathcal{X}_f$, then $z_{N+1} \in \alpha \mathcal{Z}_f$ is guaranteed by (a) and $e_{N+1} \in \mathcal{F}_N(\Phi^e)$ holds by construction. Therefore, $x_{N+1} \in \mathcal{X}_f$. $\square$

Intuitively, (a) in Lemma 2 ensures that $\alpha \mathcal{Z}_f$ is RPI with respect to control law K_f and optimized disturbance $\Gamma \bar{w}$, while (b) constrains the error states to the same tube $\mathcal{F}_N(\Phi^e)$ after N timesteps, which is less restrictive but similar to the finite impulse response (FIR) constraint in Sieber, Zanelli, et al. (2022). To use $\mathcal{X}_f$ as the terminal set in the proposed MPC, $\mathcal{X}_f$ needs to satisfy the state constraints, i.e. $\mathcal{X}_f \subseteq \mathcal{X}$, and control law (26) needs to satisfy the input constraints, i.e. $\kappa_f(x) \in \mathcal{U}$, for all $x \in \mathcal{X}_f$. This can be enforced by

$$\alpha \mathcal{Z}_f \subseteq \mathcal{X} \ominus \mathcal{F}_N(\Phi^e), \quad (28)$$

$$\alpha K_f \mathcal{Z}_f \subseteq \mathcal{U} \ominus \mathcal{F}_N(\Phi^v), \quad (29)$$

where (28) follows from $\mathcal{X}_f = \alpha \mathcal{Z}_f \oplus \mathcal{F}_N(\Phi^e)$ and (29) follows from

$$\kappa_f(x_N) = K_f \underbrace{z_N}_{\in \alpha \mathcal{Z}_f} + \underbrace{\sum_{j=0}^N \Phi_{N,j}^v \bar{w}_{N-j}}_{\in \mathcal{F}_N(\Phi^v)} \in \mathcal{U}.$$

and the matrix-valued candidates as

$$\hat{\Phi}^e = \text{shift}(\Phi^{e*}, \Phi_{N,:}^{e*}), \quad (38a)$$

$$\hat{\Phi}^v = \text{shift}(\Phi^{v*}, \Phi_{N,:}^{v*}), \quad (38b)$$

$$\hat{\Sigma} = \text{shift}(\Sigma^*, [\Xi_1^*: \sigma_N^* \cdot \mathbb{I}_n]), \quad (38c)$$

$$\hat{\Xi} = \Xi^*, \quad (38d)$$

where the $\text{shift}(\cdot, \cdot)$ operator is defined in Section 2.1, $\Phi_{N,:}$ denotes the last block row of Φ^e and Φ^v , and $\Xi_1^* = [\Xi_1^* \dots \Xi_{N-1}^*]$. The matrix-valued candidates are thus constructed by shifting the optimal solutions up and left by one block row and block column, respectively, and by appending the last block row of Φ^e , Φ^v , and $[\Xi_1^* \dots \Xi_{N-1}^* \sigma_N^* \cdot \mathbb{I}_n]$, respectively. Note that due to the choice of these candidates, we get $\hat{\sigma}_i = \sigma_{i+1}^*$, $i = 0, \dots, N-1$, $\hat{\sigma}_N = \sigma_N^*$, and $\hat{\Gamma} = \Gamma^* = A\Phi_{N,0}^{e*} + B\Phi_{N,0}^{v*} + \Xi_0^*$ by definition. Next, we show that candidates (37) fulfill nominal dynamics (32c), error dynamics (32d), and the system response constraint (32e). Using (37a) with $i = 0$, the state candidate sequence is initialized with

$$\begin{aligned} \hat{z}_0 &= z_1^* + \Phi_{1,0}^{e*} \bar{w}(k) = Az_0^* + Bv_0^* + p_0^* + \sigma_1^* \bar{w}(k) \\ &= Ax(k) + B\kappa_{\text{MPC}}(x(k)) + \eta(k) = x(k+1), \end{aligned}$$

where we used $\Phi_{1,0}^{e*} = \sigma_1^* \cdot \mathbb{I}_n$ by (36a), nominal dynamics (32c), and (35). The remainder of the state candidate sequence, i.e., $i = 1, \dots, N-1$, then trivially fulfills the nominal dynamics by definition and for the terminal candidate state, i.e. $i = N$, we get

$$\begin{aligned} \hat{z}_N &= A_{K_f} z_N^* + \Gamma^* \bar{w}(k) \\ &= A_{K_f} z_N^* + (A\Phi_{N,0}^{e*} + B\Phi_{N,0}^{v*} + \Xi_0^*) \bar{w}(k) \\ &= A(z_N^* + \Phi_{N,0}^{e*} \bar{w}(k)) + B(K_f z_N^* + \Phi_{N,0}^{v*} \bar{w}(k)) + \Xi_0^* \bar{w}(k) \\ &= A\hat{z}_{N-1} + B\hat{v}_{N-1} + \hat{p}_{N-1}, \end{aligned}$$

by definition of Γ^* and candidate sequences (37), which shows that the candidate sequences fulfill the nominal dynamics. For the error dynamics, the matrix-valued candidates (38) fulfill error dynamics (32d) and constraint (32e) by construction. Next, we need to show that candidate sequences (37) fulfill inclusion (32f). For this, we first define the tube sequences via the candidate matrices, i.e., for $i = 0, \dots, N-1$:

$$\mathcal{F}_i(\hat{\Phi}^e) = \bigoplus_{j=0}^{i-1} \hat{\Phi}_{i,j}^{e*} \bar{\mathcal{W}} = \bigoplus_{j=1}^i \Phi_{i+1,j}^{e*} \bar{\mathcal{W}}.$$

Note that above set definition together with (24) yields the following relation

$$\mathcal{F}_{i+1}(\Phi^{e*}) = \mathcal{F}_i(\hat{\Phi}^e) \oplus \Phi_{i+1,0}^{e*} \bar{\mathcal{W}}.$$

To prove that the inclusion constraint is fulfilled, we need to show that for $i = 0, \dots, N-1$

$$\hat{z}_i = z_{i+1}^* + \Phi_{i+1,0}^{e*} \bar{w}(k) \in \mathcal{X} \ominus \mathcal{F}_i(\hat{\Phi}^e). \quad (39a)$$

Given that $z_{i+1}^* \in \mathcal{X} \ominus \mathcal{F}_{i+1}(\Phi^{e*})$ and $\Phi_{i+1,0}^{e*} \bar{w}(k) \in \Phi_{i+1,0}^{e*} \bar{\mathcal{W}}$, inclusion (39a) holds for $i = 0, \dots, N-2$ with

$$\begin{aligned} &(\mathcal{X} \ominus \mathcal{F}_{i+1}(\Phi^{e*})) \oplus \Phi_{i+1,0}^{e*} \bar{\mathcal{W}} \\ &= (\mathcal{X} \ominus \mathcal{F}_i(\hat{\Phi}^e) \ominus \Phi_{i+1,0}^{e*} \bar{\mathcal{W}}) \oplus \Phi_{i+1,0}^{e*} \bar{\mathcal{W}} \subseteq \mathcal{X} \ominus \mathcal{F}_i(\hat{\Phi}^e), \end{aligned}$$

where we used Kolmanovsky and Gilbert (1998, Theorem 2.1 (ii), (v)). For the last candidate state, we need to show

$$\hat{z}_{N-1} = z_N^* + \Phi_{N,0}^{e*} \bar{w}(k) \in \mathcal{X} \ominus \mathcal{F}_{N-1}(\hat{\Phi}^e).$$

Given that $z_N^* \in \alpha^* \mathcal{Z}_f \subseteq \mathcal{X} \ominus \mathcal{F}_N(\Phi^{e*})$ and $\Phi_{N,0}^{e*} \bar{w}(k) \in \Phi_{N,0}^{e*} \bar{\mathcal{W}}$, we can confirm that

$$(\mathcal{X} \ominus \mathcal{F}_N(\Phi^{e*})) \oplus \Phi_{N,0}^{e*} \bar{\mathcal{W}} \subseteq \mathcal{X} \ominus \mathcal{F}_{N-1}(\hat{\Phi}^e).$$

The same argument is used to show that the input constraints (32g) are also fulfilled by the candidate input sequences. The terminal

set constraints (32h)–(32j) are trivially fulfilled, since the last block row of all candidate matrices is equal to the last block row of the previous optimal solution (see (38)) and $\hat{\alpha} = \alpha^*$. Finally, the constraint $\hat{z}_N \in \hat{\alpha} \mathcal{Z}_f$ in (32h) is also fulfilled since

$$\hat{z}_N = A_{K_f} z_N^* + \Gamma^* \bar{w}(k) \in \alpha^* A_{K_f} \mathcal{Z}_f \oplus \Gamma^* \bar{\mathcal{W}} \stackrel{(32j)}{\subseteq} \alpha^* \mathcal{Z}_f = \hat{\alpha} \mathcal{Z}_f.$$

To show that (32k) and (32l) hold, we first define the candidates of auxiliary variables ψ_i^d , $\psi_{i,j}^d$ for a single vertex $\mathcal{D}^d$, i.e., for $i = 0, \dots, N-2$ we have

$$\begin{aligned} \hat{\psi}_i^d &= \Delta_A^d \hat{z}_i + \Delta_B^d \hat{v}_i - \hat{p}_i \\ &= \Delta_A^d z_{i+1}^* + \Delta_B^d v_{i+1}^* - p_{i+1}^* \\ &\quad + (\Delta_A^d \Phi_{i+1,0}^{e*} + \Delta_B^d \Phi_{i+1,0}^{v*} - \Sigma_{i+2,0}^*) \bar{w}(k) \\ &= \psi_{i+1}^{d*} + \psi_{i+1,0}^{d*} \bar{w}(k), \\ \hat{\psi}_{i,j}^d &= \Delta_A^d \hat{\Phi}_{i,j}^e + \Delta_B^d \hat{\Phi}_{i,j}^v - \hat{\Sigma}_{i+1,j} \\ &= \Delta_A^d \Phi_{i+1,j+1}^{e*} + \Delta_B^d \Phi_{i+1,j+1}^{v*} - \Sigma_{i+2,j+1}^* \\ &= \psi_{i+1,j+1}^{d*}, \end{aligned}$$

and for the last timesteps, we get

$$\begin{aligned} \hat{\psi}_{N-1}^d &= \Delta_A^d \hat{z}_{N-1} + \Delta_B^d \hat{v}_{N-1} - \hat{p}_{N-1} \\ &= \Delta_A^d z_N^* + \Delta_B^d K_f z_N^* \\ &\quad + (\Delta_A^d \Phi_{N,0}^{e*} + \Delta_B^d \Phi_{N,0}^{v*} - \Xi_0^*) \bar{w}(k) \\ &= \Delta_{K_f}^d z_N^* + \psi_{N,0}^{d*} \bar{w}(k), \\ \hat{\psi}_{N-1,j}^d &= \Delta_A^d \hat{\Phi}_{N-1,j}^e + \Delta_B^d \hat{\Phi}_{N-1,j}^v - \hat{\Sigma}_{N,j} \\ &= \Delta_A^d \Phi_{N,j+1}^{e*} + \Delta_B^d \Phi_{N,j+1}^{v*} - \Xi_{j+1}^* = \psi_{N,j+1}^{d*}, \\ \hat{\psi}_{N,j}^d &= \Delta_A^d \hat{\Phi}_{N,j}^e + \Delta_B^d \hat{\Phi}_{N,j}^v - \hat{\Xi}_j \\ &= \Delta_A^d \Phi_{N,j}^{e*} + \Delta_B^d \Phi_{N,j}^{v*} - \Xi_j^* = \psi_{N,j}^{d*}. \end{aligned}$$

Using these candidates, we then check the inclusions (32k) and (32l), i.e., for $i = 0, \dots, N-2$ we get

$$\begin{aligned} \{\hat{\psi}_i^d\} \oplus \bigoplus_{j=0}^{i-1} \hat{\psi}_{i,j}^d \bar{\mathcal{W}} \oplus \mathcal{W} \\ = \{\psi_{i+1}^{d*} + \psi_{i+1,0}^{d*} \bar{w}(k)\} \oplus \bigoplus_{j=1}^i \psi_{i+1,j}^{d*} \bar{\mathcal{W}} \oplus \mathcal{W} \\ \subseteq \{\psi_{i+1}^{d*}\} \oplus \bigoplus_{j=0}^i \psi_{i+1,j}^{d*} \bar{\mathcal{W}} \oplus \mathcal{W} \stackrel{(32k)}{\subseteq} \sigma_{i+2}^* \bar{\mathcal{W}} = \hat{\sigma}_{i+1} \bar{\mathcal{W}}, \end{aligned}$$

and for $i = N-1$ we get

$$\begin{aligned} \{\hat{\psi}_{N-1}^d\} \oplus \bigoplus_{j=0}^{N-2} \hat{\psi}_{N-1,j}^d \bar{\mathcal{W}} \oplus \mathcal{W} \\ = \{\Delta_{K_f}^d z_N^* + \psi_{N,0}^{d*} \bar{w}(k)\} \oplus \bigoplus_{j=1}^{N-1} \psi_{N,j}^{d*} \bar{\mathcal{W}} \oplus \mathcal{W} \\ \subseteq \alpha^* \Delta_{K_f}^d \mathcal{Z}_f \oplus \bigoplus_{j=0}^{N-1} \psi_{N,j}^{d*} \bar{\mathcal{W}} \oplus \mathcal{W} \stackrel{(32l)}{\subseteq} \sigma_N^* \bar{\mathcal{W}} = \hat{\sigma}_N \bar{\mathcal{W}}, \end{aligned}$$

both of which satisfy (32k). Finally, (32l) is trivially fulfilled by the candidate solution, i.e.,

$$\begin{aligned} \hat{\alpha} \Delta_{K_f}^d \mathcal{Z}_f \oplus \bigoplus_{j=0}^{N-1} \hat{\psi}_{N,j}^d \bar{\mathcal{W}} \oplus \mathcal{W} \\ = \alpha^* \Delta_{K_f}^d \mathcal{Z}_f \oplus \bigoplus_{j=0}^{N-1} \psi_{N,j}^{d*} \bar{\mathcal{W}} \oplus \mathcal{W} \stackrel{(32l)}{\subseteq} \sigma_N^* \bar{\mathcal{W}} = \hat{\sigma}_N \bar{\mathcal{W}}. \end{aligned}$$

Since we impose (32k) and (32l) for each vertex $\mathcal{D}^d$ individually, the argument holds for all vertices of $\mathcal{D}$. Therefore, we have shown that the candidate solutions (37), (38) satisfy all constraints in (32), which proves recursive feasibility. $\square$

Next, we show that the closed-loop system is input-to-state stable (ISS), where ISS is formally defined in Rawlings et al. (2019, Definition B.42). For the ISS proof, we rely on an ISS-Lyapunov function (Rawlings et al., 2019, Appendix B.6), which we define in the following definition. Note that due to the equivalence of (1) and (35), we only need to check the stability of (35) to conclude ISS of the original system.

Definition 3 (ISS-Lyapunov Function). Consider an RPI set $\mathcal{S} \subset \mathbb{R}^n$ with the origin in its interior. A function $V : \mathcal{S} \rightarrow \mathbb{R}_{\geq 0}$ is called an ISS-Lyapunov function in $\mathcal{S}$ for system (35) with $u = \kappa_{\text{MPC}}(x)$ and $\bar{w} \in \bar{\mathcal{W}}$, if there exist $\mathcal{K}$ -functions γ_1, γ_2 , and γ_3 , such that for all $x \in \mathcal{S}$ and $\bar{w} \in \bar{\mathcal{W}}$:

$$\gamma_1(\|x\|) \leq V(x) \leq \gamma_2(\|x\|), \quad (40a)$$

$$V(x^+) - V(x) \leq -\gamma_1(\|x\|) + \gamma_3(\|\bar{w}\|), \quad (40b)$$

where $\|\cdot\|$ is any vector norm on $\mathbb{R}^n$.

Theorem 4. Given that Assumption 1 holds, system (1) subject to constraints (3) and in closed-loop with (33), is input-to-state stable (ISS) in $\mathcal{X}_{\text{feas}}$ for any admissible sequence of combined disturbances η , where $\mathcal{X}_{\text{feas}}$ is the set of all initial states $x(k)$ for which (32) is feasible.

Proof. Due to Theorem 3, we know that (32) is recursively feasible for all $x(k) \in \mathcal{X}_{\text{feas}}$. Therefore, we can prove ISS by showing that the optimal value function of (32) is an ISS-Lyapunov function in $\mathcal{X}_{\text{feas}}$ according to Definition 3 (Rawlings et al., 2019, Lemma B.44). Denote (32a) as $V_N(x(k); \mathbf{z}, \mathbf{v})$ and the optimal value function of (32) as $V_N^*(x(k))$, i.e., $V_N(\cdot)$ evaluated at the optimal solution $\mathbf{z}^*, \mathbf{v}^*$. Using the candidate sequences $\hat{\mathbf{z}}, \hat{\mathbf{v}}$ defined in (37), it follows that

$$\begin{aligned} V_N^*(x(k+1)) - V_N^*(x(k)) &\leq V_N(x(k+1); \hat{\mathbf{z}}, \hat{\mathbf{v}}) - V_N^*(x(k)) \\ &= l_f(\hat{\mathbf{z}}_N) - l_f(\mathbf{z}_N^*) + \sum_{i=0}^{N-1} l(\hat{\mathbf{z}}_i, \hat{\mathbf{v}}_i) - l(\mathbf{z}_i^*, \mathbf{v}_i^*). \end{aligned}$$

We then use the candidate sequences (37) and uniform continuity of $l_f(\cdot)$ with corresponding $\mathcal{K}$ -function $\beta_f(\cdot)$ to obtain the upper bound

$$\begin{aligned} l_f(\hat{\mathbf{z}}_N) - l_f(\mathbf{z}_N^*) &= l_f(A_{K_f} \mathbf{z}_N^* + \Gamma^* \bar{w}(k)) - l_f(\mathbf{z}_N^*) \\ &\quad + l_f(A_{K_f} \mathbf{z}_N^*) - l_f(A_{K_f} \mathbf{z}_N^*) \\ &\leq l_f(A_{K_f} \mathbf{z}_N^*) - l_f(\mathbf{z}_N^*) \\ &\quad + \|l_f(A_{K_f} \mathbf{z}_N^* + \Gamma^* \bar{w}(k)) - l_f(A_{K_f} \mathbf{z}_N^*)\| \\ &\leq l_f(A_{K_f} \mathbf{z}_N^*) - l_f(\mathbf{z}_N^*) + \beta_f(\|\Gamma^* \bar{w}(k)\|). \end{aligned}$$

We rewrite the sum of the cost differences as

$$\begin{aligned} \sum_{i=0}^{N-1} l(\hat{\mathbf{z}}_i, \hat{\mathbf{v}}_i) - l(\mathbf{z}_i^*, \mathbf{v}_i^*) &= -l(\mathbf{z}_0^*, \mathbf{v}_0^*) + l(\hat{\mathbf{z}}_{N-1}, \hat{\mathbf{v}}_{N-1}) \\ &\quad + \sum_{i=1}^{N-1} l(\hat{\mathbf{z}}_{i-1}, \hat{\mathbf{v}}_{i-1}) - l(\mathbf{z}_i^*, \mathbf{v}_i^*), \end{aligned}$$

and use the candidate sequences (37) and uniform continuity of $l(\cdot, \cdot)$ with $\beta_x(\cdot), \beta_u(\cdot)$ the respective $\mathcal{K}$ -functions to obtain

$$\begin{aligned} l(\hat{\mathbf{z}}_{i-1}, \hat{\mathbf{v}}_{i-1}) - l(\mathbf{z}_i^*, \mathbf{v}_i^*) &= l(\mathbf{z}_i^* + \Phi_{i,0}^{e*} \bar{w}(k), \mathbf{v}_i^* + \Phi_{i,0}^{v*} \bar{w}(k)) \\ &\quad - l(\mathbf{z}_i^*, \mathbf{v}_i^*) \\ &= l(\mathbf{z}_i^* + \Phi_{i,0}^{e*} \bar{w}(k), \mathbf{v}_i^* + \Phi_{i,0}^{v*} \bar{w}(k)) \\ &\quad - l(\mathbf{z}_i^*, \mathbf{v}_i^* + \Phi_{i,0}^{v*} \bar{w}(k)) \\ &\quad + l(\mathbf{z}_i^*, \mathbf{v}_i^* + \Phi_{i,0}^{v*} \bar{w}(k)) - l(\mathbf{z}_i^*, \mathbf{v}_i^*) \\ &\leq \beta_x(\|\Phi_{i,0}^{e*} \bar{w}(k)\|) + \beta_u(\|\Phi_{i,0}^{v*} \bar{w}(k)\|). \end{aligned}$$

Additionally, we rewrite

$$\begin{aligned} l(\hat{\mathbf{z}}_{N-1}, \hat{\mathbf{v}}_{N-1}) &= l(\mathbf{z}_N^* + \Phi_{N,0}^{e*} \bar{w}(k), K_f \mathbf{z}_N^* + \Phi_{N,0}^{v*} \bar{w}(k)) \\ &= l(\mathbf{z}_N^* + \Phi_{N,0}^{e*} \bar{w}(k), K_f \mathbf{z}_N^* + \Phi_{N,0}^{v*} \bar{w}(k)) \\ &\quad - l(\mathbf{z}_N^*, K_f \mathbf{z}_N^* + \Phi_{N,0}^{v*} \bar{w}(k)) \\ &\quad + l(\mathbf{z}_N^*, K_f \mathbf{z}_N^* + \Phi_{N,0}^{v*} \bar{w}(k)) \\ &\quad - l(\mathbf{z}_N^*, K_f \mathbf{z}_N^*) + l(\mathbf{z}_N^*, K_f \mathbf{z}_N^*) \\ &\leq \beta_x(\|\Phi_{N,0}^{e*} \bar{w}(k)\|) + \beta_u(\|\Phi_{N,0}^{v*} \bar{w}(k)\|) \\ &\quad + l(\mathbf{z}_N^*, K_f \mathbf{z}_N^*), \end{aligned}$$

and by combining the above, we get the value function decrease

$$\begin{aligned} V_N^*(x(k+1)) - V_N^*(x(k)) &\leq -l(\mathbf{z}_0^*, \mathbf{v}_0^*) + l(\mathbf{z}_N^*, K_f \mathbf{z}_N^*) \\ &\quad + l_f(A_{K_f} \mathbf{z}_N^*) - l_f(\mathbf{z}_N^*) \\ &\quad + \gamma_3(\|\bar{w}(k)\|), \end{aligned}$$

where we collected all $\mathcal{K}$ -functions in

$$\begin{aligned} \gamma_3(\|\bar{w}(k)\|) &= \beta_f(\|\Gamma^* \bar{w}(k)\|) \\ &\quad + \sum_{i=1}^N \beta_x(\|\Phi_{i,0}^{e*} \bar{w}(k)\|) + \beta_u(\|\Phi_{i,0}^{v*} \bar{w}(k)\|). \end{aligned} \quad (41)$$

Due to compactness of $\mathcal{X}$ and $\mathcal{U}$, the variables $\Phi_{i,0}^{e*}, \Phi_{i,0}^{v*}$, and Γ^* are bounded. Therefore, (41) is a sum of compositions of $\mathcal{K}$ -functions, e.g. $\beta_f(\|\Gamma^* \bar{w}(k)\|) \leq \beta_f(\|\Gamma^*\| \|\bar{w}(k)\|)$, which we can rewrite as a single $\mathcal{K}$ -function $\gamma_3(\cdot)$. Finally, we notice that $-l(\mathbf{z}_0^*, \mathbf{v}_0^*) = -l(x(k), \kappa_{\text{MPC}}(x(k))) \leq -\gamma_1(\|x(k)\|)$ and $l_f(A_{K_f} \mathbf{z}_N^*) - l_f(\mathbf{z}_N^*) \leq -l(\mathbf{z}_N^*, K_f \mathbf{z}_N^*)$ due to Assumption 1 and therefore obtain

$$V_N^*(x(k+1)) - V_N^*(x(k)) \leq -\gamma_1(\|x(k)\|) + \gamma_3(\|\bar{w}(k)\|).$$

Since $\gamma_3(\cdot)$ is a $\mathcal{K}$ -function, $V_N^*(\cdot)$ fulfills decrease condition (40b) and because $V_N^*(x) \geq l(x, \kappa_{\text{MPC}}(x)) \forall x \in \mathcal{X}_{\text{feas}}$, the optimal value function is lower bounded by the $\mathcal{K}$ -function $\gamma_1(\cdot)$ by assumption, i.e., $V_N^*(x) \geq \gamma_1(\|x\|) \forall x \in \mathcal{X}_{\text{feas}}$. Using the facts that $l_f(\cdot)$ is upper bounded, the state and input constraints are compact, and monotonicity of $V_N^*(\cdot)$, we can conclude that $V_N^*(\cdot)$ is upper bounded by a $\mathcal{K}$ -function $\gamma_2(\cdot)$ in $\mathcal{X}_{\text{feas}}$ due to Rawlings et al. (2019, Propositions 2.15 – 2.16). Therefore, $V_N^*(\cdot)$ also fulfills (40a) and is thus an ISS-Lyapunov function in $\mathcal{X}_{\text{feas}}$ (Definition 3). Hence, we have shown ISS of system (35) in closed-loop with (33), which implies ISS of the original system (1) with SLTMPC controller (33). $\square$

Remark 4. Contrary to Limon et al. (2009), Raimondo, Limon, Lazar, Magni, and Fernández Camacho (2009), we do not treat the components of the combined uncertainty $\eta = d + w$ separately, where the state dependent model uncertainty d is assumed to have a stability margin that is incorporated in $-\gamma_1(\|x\|)$ (Raimondo et al., 2009, Remark 3). Instead we use (35) with a single disturbance and the standard ISS proof technique (Rawlings et al., 2019, Appendix B.6). Intuitively, we treat (35) as

$$x^+ = Ax + \begin{bmatrix} B & I_n \end{bmatrix} \begin{bmatrix} v_0^* \\ p_0^* \end{bmatrix} + \sigma_1^* w(k)$$

Table 1

Computational complexity of the two proposed SLTMPC methods (23), (32) and the method proposed in Chen et al. (2024) in terms of optimization variables and constraints. The dimensions used below, e.g. n_x , are defined in Section 2.

	Number of variables	Number of constraints
Chen et al. (2024)	$\mathcal{O}(N^2(n^2 + nm + n_x n_{\bar{w}} + n_u n_{\bar{w}}) + N(n_f n_{\bar{w}}))$	$\mathcal{O}(N^2 n_{\bar{w}}(n_x + n_u) + N(n + n_f n_{\bar{w}}))$
SLTMPC (23)	$\mathcal{O}(N^2(n^2 + nm + n_x n_{\bar{w}} + n_u n_{\bar{w}}) + N(n_f n_{\bar{w}} + n_y^2 n_D))$	$\mathcal{O}(N^2 n_{\bar{w}}(n_x + n_u) + N(n + n_f n_{\bar{w}} + n_y^2 n_D))$
Rec. Feas. SLTMPC (32)	$\mathcal{O}(N^2(n^2 + nm + n_x n_{\bar{w}} + n_u n_{\bar{w}}) + N(n_f n_{\bar{w}} + n_y^2 n_D + n^2) + n_f^2)$	$\mathcal{O}(N^2 n_{\bar{w}}(n_x + n_u) + (N + 1)(n + n_f n_{\bar{w}} + n_y^2 n_D) + n_f^2)$

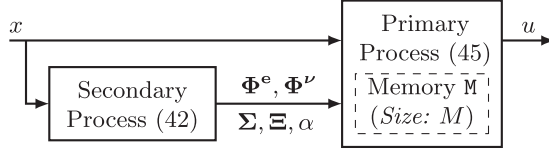


Fig. 1. Visualization of the asynchronous computation scheme for filter-based SLTMPC: the secondary process computes the tube parameters Φ^e , Φ^v , Σ , Ξ , α , which are then passed to the primary process and stored in memory M .

in the proof, where $[v_0^{*T} p_0^{*T}]^T$ is an extended input and σ_1^* is an optimized but bounded disturbance gain. Also note that $\Phi_{i,0}^{e*}$, $\Phi_{i,0}^{v*}$, and Γ^* in (41) are optimization variables that vary each time (32) is solved. Therefore, they affect the size of the region around the origin where the closed-loop system converges to. Adding a regularizer on these variables thus allows tuning of this convergence region.

5. SLTMPC with asynchronous updates

Filter-based SLTMPC methods (23), (32), and Chen et al. (2024) are computationally demanding due to the concurrent optimization of nominal trajectories $\mathbf{z}$, $\mathbf{v}$, $\mathbf{p}$, error dynamics (15) parameterized by error system responses Φ^e , Φ^v and disturbance filter Σ , Ξ , and terminal set scaling α . The computational complexity of these methods is stated in Table 1. This complexity renders the methods unsuitable for practical applications with fast sampling times and limited computational resources. To alleviate some of the computational burden, we extend the asynchronous computation scheme introduced in Sieber et al. (2023) for filter-based SLTMPC. To reduce the complexity of (32), we remove error dynamics parameters Φ^e , Φ^v , Σ , Ξ , and terminal scaling α as optimization variables and instead use values that are computed at a lower frequency. The asynchronous computation scheme uses this idea to split the optimization of (32) into two separate processes – the primary and the secondary process. The secondary process computes the error dynamics parameters and the terminal set scaling, before passing them to the primary process, which only optimizes the nominal trajectories $\mathbf{z}$, $\mathbf{v}$, $\mathbf{p}$. The asynchronous computation scheme therefore enables running the primary process in closed-loop with system (1) at a high frequency. The computationally expensive secondary process then runs concurrently at a lower frequency and reduces conservativeness by periodically providing new error dynamics parameters and terminal set scalings. Apart from limited computational resources, practical applications also impose memory constraints, i.e., we can only store a limited number of error dynamics parameters and terminal set scalings, which we handle by specific memory update rules. Fig. 1 provides an overview of this scheme.

5.1. Secondary process: Optimize tubes

The main objective of the secondary process is to compute error dynamics parameters Φ^e , Φ^v , Σ , Ξ , and terminal set scaling α , which define the state, input, and disturbance tubes, and the terminal set. Therefore, we can choose the cost function

Algorithm 1 Update memory M (Secondary Process)

Input: M_{new} , $M(k-1)$, $\lambda(k-1)$
Output: $M(k)$

```

1: procedure UPDATEMEMORY( $M_{\text{new}}$ )
2:   ▷ runs when the secondary process terminates
3:    $M(k) \leftarrow M(k-1)$ 
4:   if  $M(k)$  not full then
5:      $m \leftarrow$  index of first empty memory slot
6:      $M_m(k) \leftarrow M_{\text{new}}$ 
7:   else
8:     ▷ determine a memory slot (except  $m = 0$ )
9:      $m \leftarrow F(M(k))$ 
10:     $M_m(k) \leftarrow M_{\text{new}}$ 

```

of the secondary process according to any criterion the tubes and the terminal set should fulfill, e.g. a cost on Φ^e promotes more aggressive tube controllers, thus leading to smaller error bounds on the state trajectories. For more details on possible cost functions and their effect on the tubes we refer to Sieber et al. (2023, Section 3.2). We formulate the secondary process similarly to (32) but with a modified objective, i.e.,

$$\min_{\Phi^e, \Phi^v, \Sigma, \Xi, \alpha \geq 0} L(\Phi^e, \Phi^v, \Sigma, \Xi), \quad (42a)$$

$$\text{s.t. (32b)–(32I),} \quad (42b)$$

where $L(\cdot)$ is the chosen cost function. After termination of (42), we use the optimized error system responses Φ^e , Φ^v to compute the tightened state and input constraints for $i = 0, \dots, N$ as

$$\mathcal{Z}_i = \mathcal{X} \ominus \mathcal{F}_i(\Phi^e), \quad \mathcal{V}_i = \mathcal{U} \ominus \mathcal{F}_i(\Phi^v). \quad (43)$$

Additionally, we use the optimized disturbance filter Σ , Ξ to compute the disturbance tubes for $d = 1, \dots, n_D$ and $i = 0, \dots, N - 1$ as

$$\mathcal{Q}_i^d = \sigma_{i+1} \bar{\mathcal{V}} \ominus \mathcal{F}_i(\Psi^d), \quad \text{with } \mathcal{F}_i(\Psi^d) := \mathcal{W} \oplus \bigoplus_{j=0}^{i-1} \Psi_{i,j}^d \bar{\mathcal{V}},$$

and $\Psi_{i,j}^d$ defined in (18). For $i = N$, the disturbance tubes are computed as

$$\mathcal{Q}_N^d = \sigma_N \bar{\mathcal{V}} \ominus \mathcal{F}_N(\Psi^d). \quad (44)$$

These sets are then passed to the primary process together with the error system responses, the disturbance filter, and the terminal set scaling as $M_{\text{new}} = (\mathcal{Z}_i, \mathcal{V}_i, \mathcal{Q}_i^d, \alpha, \Phi^e, \Phi^v, \Sigma, \Xi)$. Since the memory $M = \{M_m \mid m = 0, \dots, M-1\}$ – where we denote the m th memory entry as M_m – is finite, we need to define a procedure that updates the memory whenever the secondary process has computed a new M_{new} , thus rendering the memory time-varying. In case the memory is full, the proposed update procedure in Algorithm 1 selects a memory slot M_m to store M_{new} according to a heuristic $F(M(k))$, e.g., choosing the least important slot for the previous solution of the primary process.

5.2. Primary process: Optimize nominal trajectory

Since a selection of tightened state and input constraints, disturbance tubes, and terminal sets is stored in memory M ,

Algorithm 2 Update memory M (Primary Process)

Input: $M_{\text{prev}}, M(k-1)$
Output: $M(k)$

1: **procedure** UPDATEMEMORY(M_{prev})
2: ▷ runs when the primary process terminates
3: $M(k) \leftarrow M(k-1)$
4: $M_0(k) \leftarrow M_{\text{prev}}$ where M_{prev} is given by (50)

the primary process can use this information to optimize the nominal trajectories. However, rather than using a single memory entry M_m , all entries are *fused* using a convex combination similar to Kögel and Findeisen (2020), Sieber et al. (2023), ensuring all available information is optimally used. The primary process is formulated as

$$\min_{\mathbf{z}, \mathbf{v}, \mathbf{p}, \boldsymbol{\lambda}} \quad l_f(z_N) + \sum_{i=0}^{N-1} l(z_i, v_i), \quad (45a)$$

$$\text{s.t. } \forall i = 0, \dots, N-1:$$

$$z_0 = x(k), \quad (45b)$$

$$z_{i+1} = Az_i + Bv_i + p_i, \quad (45c)$$

$$z_i \in \bigoplus_{m=0}^{M-1} \lambda_m \mathcal{Z}_i^m, \quad (45d)$$

$$v_i \in \bigoplus_{m=0}^{M-1} \lambda_m \mathcal{V}_i^m, \quad (45e)$$

$$z_N \in \bigoplus_{m=0}^{M-1} \lambda_m \alpha^m \mathcal{Z}_f, \quad (45f)$$

$$\psi_i^d \in \bigoplus_{m=0}^{M-1} \lambda_m \mathcal{Q}_i^{d,m}, \quad \forall d = 1, \dots, n_D, \quad (45g)$$

$$\lambda_m \geq 0, \quad \sum_{m=0}^{M-1} \lambda_m = 1, \quad (45h)$$

where $\boldsymbol{\lambda} = [\lambda_0 \dots \lambda_{M-1}]$ are the convex combination variables, the costs $l_f(\cdot)$, $l(\cdot, \cdot)$ fulfill Assumption 1, ψ_i^d is defined as in (18), and $\mathcal{Z}_i^m$, $\mathcal{V}_i^m$, $\mathcal{Q}_i^{d,m}$, α^m are retrieved from memory slot M_m . The resulting MPC control law is given by

$$\kappa_{\text{primary}}(x(k)) = v_0^*, \quad (46)$$

where v_0^* is the first element of optimizer $\mathbf{v}^*$ of (45).

Since the tubes computed by the secondary process are more general than those used in Sieber et al. (2023), we cannot use the proof in Sieber et al. (2023) to show recursive feasibility of the primary process. Therefore, we additionally store the previous solution of (45) in memory slot M_0 as a fallback solution. Specifically, we store the shifted convex combination of tubes after the primary process terminates, i.e., for all $d = 1, \dots, n_D$ and $i = 0, \dots, N-1$ we compute

$$\mathcal{Z}_i^{\text{prev}} = \bigoplus_{m=0}^{M-1} \lambda_m^* (\mathcal{X} \ominus \mathcal{F}_i(\Phi_{\text{shift}}^{\mathbf{e},m})), \quad (47a)$$

$$\mathcal{V}_i^{\text{prev}} = \bigoplus_{m=0}^{M-1} \lambda_m^* (\mathcal{U} \ominus \mathcal{F}_i(\Phi_{\text{shift}}^{\mathbf{v},m})), \quad (47b)$$

$$\mathcal{Q}_i^{d,\text{prev}} = \bigoplus_{m=0}^{M-1} \lambda_m^* (\sigma_{i+2}^m \bar{\mathcal{W}} \ominus \mathcal{F}_i(\Psi_{\text{shift}}^{d,m})), \quad (47c)$$

where the shifted system responses are computed as

$$\Phi_{\text{shift}}^{\mathbf{e},m} = \text{shift}(\Phi^{\mathbf{e},m}, \Phi_{N,:}^{\mathbf{e},m}), \quad (48a)$$

$$\Phi_{\text{shift}}^{\mathbf{v},m} = \text{shift}(\Phi^{\mathbf{v},m}, \Phi_{N,:}^{\mathbf{v},m}), \quad (48b)$$

with the $\text{shift}(\cdot, \cdot)$ operator defined in Section 2.1 and $\Phi_{N,:}$ denoting the last block row of $\Phi^{\mathbf{e},m}$ and $\Phi^{\mathbf{v},m}$. The shifted auxiliary variable $\Psi_{\text{shift}}^{d,m}$ is computed with (18) using the shifted system responses and the shifted disturbance filter, i.e.,

$$\Sigma_{\text{shift}}^m = \text{shift}(\Sigma^m, [\Xi_1^m \sigma_N^m \cdot \mathbb{I}_n]), \quad (49a)$$

$$\Xi_{\text{shift}}^m = \Xi^m, \quad (49b)$$

where $\Xi^m = [\Xi_1^m \dots \Xi_{N-1}^m]$ and the σ_i are obtained from either $\Phi^{\mathbf{e},m}$ or Σ^m using (36a). Additionally, we compute the tubes for $i = N$ as

$$\mathcal{Z}_N^{\text{prev}} = \bigoplus_{m=0}^{M-1} \lambda_m^* \mathcal{Z}_N^m, \quad \mathcal{V}_N^{\text{prev}} = \bigoplus_{m=0}^{M-1} \lambda_m^* \mathcal{V}_N^m, \\ \mathcal{Q}_N^{d,\text{prev}} = \bigoplus_{m=0}^{M-1} \lambda_m^* \mathcal{Q}_N^{d,m}.$$

Apart from the tubes, we also store the fused terminal set scaling, the shifted error system responses, and the shifted disturbance filter, since they are needed to recompute (47) in case $\lambda_0^* \neq 0$ in the next solution of (45). Therefore, we compute $\alpha^{\text{prev}} = \sum_{m=0}^{M-1} \lambda_m^* \alpha^m$ and

$$\Phi^{\mathbf{e},\text{prev}} = \sum_{m=0}^{M-1} \lambda_m^* \Phi_{\text{shift}}^{\mathbf{e},m}, \quad \Phi^{\mathbf{v},\text{prev}} = \sum_{m=0}^{M-1} \lambda_m^* \Phi_{\text{shift}}^{\mathbf{v},m}, \\ \Sigma^{\text{prev}} = \sum_{m=0}^{M-1} \lambda_m^* \Sigma_{\text{shift}}^m, \quad \Xi^{\text{prev}} = \sum_{m=0}^{M-1} \lambda_m^* \Xi_{\text{shift}}^m,$$

before collecting all computed sets and parameters in memory entry

$$M_{\text{prev}} = (\mathcal{Z}_i^{\text{prev}}, \mathcal{V}_i^{\text{prev}}, \mathcal{Q}_i^{d,\text{prev}}, \alpha^{\text{prev}}, \\ \Phi^{\mathbf{e},\text{prev}}, \Phi^{\mathbf{v},\text{prev}}, \Sigma^{\text{prev}}, \Xi^{\text{prev}}), \quad (50)$$

and updating the memory according to Algorithm 2.

5.3. Theoretical guarantees

In the following, we provide theoretical guarantees for the proposed asynchronous computation scheme applied to (1), i.e., we prove recursive feasibility and ISS of primary process (45) for any state of the memory, including the case in which the memory is updated. For the proofs, we make use of the following assumption.

Assumption 2. At the start of the control task, memory M contains at least one memory entry M_0 for which the primary process is feasible, i.e., (45) is feasible for $M_0 = (\mathcal{Z}_i^0, \mathcal{V}_i^0, \mathcal{Q}_i^{d,0}, \alpha^0, \Phi^{\mathbf{e},0}, \Phi^{\mathbf{v},0}, \Sigma^0, \Xi^0)$ with $\lambda_0 = 1$.

Note that Assumption 2 is satisfied, if for example the secondary process is once run offline and the memory is initialized with the computed memory entry M_{new} .

Proposition 5. Let Assumption 2 hold. Then, primary process (45) is recursively feasible for any state of memory $M(k)$ and any memory updates according to Algorithms 1 and 2.

Proof. We prove the proposition using the standard shifting argument (Rawlings et al., 2019) and relying on the proof of

Theorem 3. We construct the candidate sequences specifically for the case in which we only use memory slot M_0 that contains the shifted previous tubes (50). This allows us to show recursive feasibility for any state of the memory, including the case in which the memory is updated, since we can always fallback to the tubes in M_0 . Due to Assumption 2, this then guarantees recursive feasibility. Let $\mathbf{z}^* = [z_0^*, \dots, z_N^*]^\top$, $\mathbf{v}^* = [v_0^*, \dots, v_{N-1}^*]^\top$, $\mathbf{p}^* = [p_0^*, \dots, p_{N-1}^*]^\top$, and $\boldsymbol{\lambda}^* = [\lambda_0^*, \dots, \lambda_{M-1}^*]^\top$ denote the optimal solution of (45) for initial state $\mathbf{x}(k)$. We then use the candidate $\hat{\lambda}_0 = 1$, $\hat{\lambda}_m = 0$, $m = 1, \dots, M-1$ for the convex combination variables and construct the remaining candidate sequences as

$$\hat{z}_i = z_{i+1}^* + \sum_{m=0}^{M-1} \lambda_m^* \Phi_{i+1,0}^{e,m} \bar{w}(k), \quad i = 0, \dots, N-1, \quad (51a)$$

$$\hat{z}_N = A_{K_f} z_N^* + \sum_{m=0}^{M-1} \lambda_m^* \Gamma^m \bar{w}(k), \quad (51b)$$

$$\hat{v}_i = v_{i+1}^* + \sum_{m=0}^{M-1} \lambda_m^* \Phi_{i+1,0}^{v,m} \bar{w}(k), \quad i = 0, \dots, N-2, \quad (51c)$$

$$\hat{v}_{N-1} = K_f z_N^* + \sum_{m=0}^{M-1} \lambda_m^* \Phi_{N,0}^{v,m} \bar{w}(k), \quad (51d)$$

$$\hat{p}_i = p_{i+1}^* + \sum_{m=0}^{M-1} \lambda_m^* \Sigma_{i+2,0}^m \bar{w}(k), \quad i = 0, \dots, N-2, \quad (51e)$$

$$\hat{p}_{N-1} = \sum_{m=0}^{M-1} \lambda_m^* \Sigma_0^m \bar{w}(k), \quad (51f)$$

with $\Gamma^m = A\Phi_{N,0}^{e,m} + B\Phi_{N,0}^{v,m} + \Sigma_0^m$. Next, we show that candidates (51) satisfy all constraints using the proof of Theorem 3. Note that (51) are equivalent to the vector-valued candidates in (37) and the tubes stored in M_0 are constructed equivalently to (38), apart from the convex combination of the system responses and disturbance filters. However, since the candidates and the tubes use the same convex combination, we can treat each contribution to the convex combination individually. For example, to show $\hat{z}_i \in \mathcal{Z}_i^0$, $i = 0, \dots, N-1$, we get

$$\sum_{m=0}^{M-1} \lambda_m^* (z_{i+1}^* + \Phi_{i+1,0}^{e,m} \bar{w}(k)) \in \bigoplus_{m=0}^{M-1} \lambda_m^* (\mathcal{X} \ominus \mathcal{F}_i(\Phi_{i+1,0}^{e,m})),$$

and we can show $z_{i+1}^* + \Phi_{i+1,0}^{e,m} \bar{w}(k) \in \mathcal{X} \ominus \mathcal{F}_i(\Phi_{i+1,0}^{e,m})$ for each m individually. As shown in the proof of Theorem 3, tubes (47) satisfy this inclusion by construction. The same argument holds for the other constraints and thus we have shown that candidate (51) satisfies all constraints in (45), which proves recursive feasibility. $\square$

For robust stability, we note that costs $l(\cdot)$, $l_f(\cdot)$ in (45a) are only functions of the nominal trajectories and independent of the convex combination variables. Therefore, the following stability result is a direct consequence of Theorem 4 and Proposition 5.

Corollary 6. Given that Assumption 1 holds, system (1) subject to constraints (3) and in closed-loop with (46), is ISS in the feasible set $\mathcal{X}_{\text{feas}}$ for any admissible sequence of combined disturbances $\boldsymbol{\eta}$.

6. Numerical results

We first compare recursively feasible SLTMPC (32) to the method presented in Chen et al. (2024) and standard tube-based MPC methods from the literature Bujarbaruah et al. (2021, 2022), Köhler et al. (2019), Langson et al. (2004), Lorenzen et al. (2019),

Lu and Cannon (2019) on a two dimensional double integrator example, before showing the benefits of the asynchronous computation scheme on a more complex example. All examples are implemented in Python using CVXPY (Diamond & Boyd, 2016) and are solved using MOSEK (MOSEK ApS, 2021). The examples were run on a machine equipped with a [3.1]GHz CPU and [32]GB of RAM.

6.1. Double integrator

In the first example, we use the same system as in Chen et al. (2024), i.e., uncertain LTI system (1) with discrete-time dynamic matrices

$$A = \begin{bmatrix} 1 & 0.15 \\ 0.1 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 0.1 \\ 1.1 \end{bmatrix},$$

subject to polytopic constraints $\|\mathbf{x}\|_\infty \leq 8$, $\|\mathbf{u}\|_\infty \leq 4$, additive disturbance $\|w\|_\infty \leq \sigma_w$, and model uncertainty

$$\Delta_A \in \text{co} \left\{ \begin{bmatrix} \epsilon_A & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} -\epsilon_A & 0 \\ 0 & 0 \end{bmatrix} \right\}, \quad \Delta_B \in \text{co} \left\{ \begin{bmatrix} 0 \\ \epsilon_B \end{bmatrix}, \begin{bmatrix} 0 \\ -\epsilon_B \end{bmatrix} \right\},$$

where the uncertainty parameters σ_w , ϵ_A , and ϵ_B are varied. Additionally, we use the cost function $l(z, v) = z^\top Q z + v^\top R v$ with $Q = 10 \cdot \mathbb{I}_2$ and $R = 1$, and horizon $N \in \{5, 10\}$. As the terminal sets, we use the maximal robust control invariant (RCI) set of (1) for Chen et al. (2024) and the maximal RPI set (Definition 2) – with K_f the LQR controller – for (32). We design the auxiliary disturbance set $\bar{\mathcal{W}}$ to be equivalent to $\mathcal{W}$ for both methods and since these are hyperrectangles we use Remark 3 to modify (32). The setup of the other methods is equivalent to Chen et al. (2024, Section 6) and we refer to Chen et al. (2024) for all details.

Fig. 2 shows the region of attraction (RoA), i.e., the set of initial states for which the MPC method is feasible, for all considered methods as a fraction of the maximal RCI set for dynamics (1). We vary the parametric uncertainty $\epsilon_A \in [0.0, 0.5]$, while fixing $\epsilon_B = \sigma_w = 0.1$ (left) and vary the additive disturbance $\sigma_w \in [0.0, 0.6]$, while fixing $\epsilon_A = \epsilon_B = 0.1$ (right). We notice that both the methods in Chen et al. (2024) and (32) outperform the other methods, especially for larger uncertainties. Generally, the method in Chen et al. (2024) is less conservative than (32), i.e., its RoA covers a larger part of the maximal RCI set. This is due to the additional constraints in (32) needed for recursive feasibility and the use of an RCI terminal set compared to the RPI terminal set in (32). However, with increasing horizon N the RoAs of the two methods become more similar. This is because the method in Chen et al. (2024) tightens the RCI terminal set with the disturbance overapproximation, which becomes more conservative for larger horizons, thus reducing the RoA. In contrast, the RoA of (32) increases with the horizon, since the MPC controller is enhancing the terminal RPI set, thus reducing conservativeness with larger horizons. The average computation times for the RoA computations are stated in Table 2. To analyze the choice of $\bar{\mathcal{W}}$ on the performance, we compare the method in Chen et al. (2024) to (32) for the same setup as in Fig. 2 (right) but with a different uncertainty description $\mathcal{W}$, i.e. (2a) with $h_w = (0.5\sigma_w, \sigma_w, 0.5\sigma_w, 0.5\sigma_w, \sigma_w, 0.5\sigma_w)$. While $\bar{\mathcal{W}}$ is fixed for Chen et al. (2024), we choose $\bar{\mathcal{W}}$ once as a hyperrectangle $\bar{\mathcal{W}}_{\text{box}}$ and once as $\bar{\mathcal{W}} = \mathcal{W}$ for (32). The results are shown in Fig. 3, where we note that (32) is less conservative than the method in Chen et al. (2024), since the chosen $\bar{\mathcal{W}}$ enable better overapproximations of the system uncertainties $\mathcal{W}$, $\mathcal{D}$. Additionally, the method in Chen et al. (2024) has significant limitations, i.e., it requires a shrinking horizon implementation and a RPI set for dynamics (1), both of which are highlighted in the following. First, we compare the closed-loop performance of the two methods. Fig. 4 shows 200 closed-loop state and input trajectories starting in $\mathbf{x}(0) =$

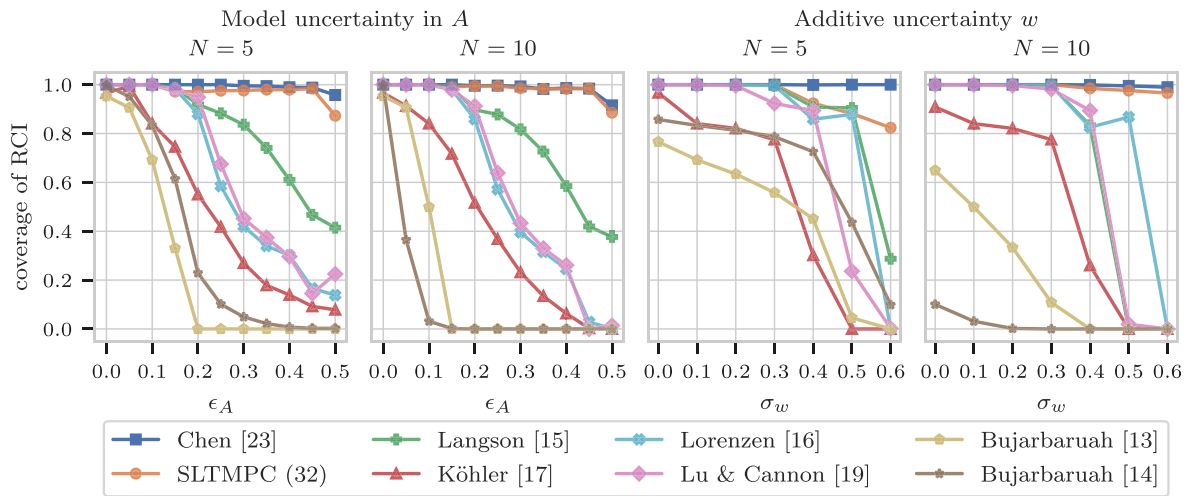


Fig. 2. Comparison of (32) to Chen et al. (2024) and other standard tube-based MPC methods (Bujarbaruah et al., 2021, 2022; Köhler et al., 2019; Langson et al., 2004; Lorenzen et al., 2019; Lu & Cannon, 2019): Region of attraction (RoA) relative to the maximal RCI set in percent, where we vary the uncertainty in the dynamics matrix A , i.e. ϵ_A , (left) and the additive disturbance, i.e. σ_w , (right). The other uncertainty parameters are fixed to $\epsilon_B = 0.1$, $\sigma_w = 0.1$ and $\epsilon_A = 0.1$, $\epsilon_B = 0.1$, respectively, and the horizon is $N = \{5, 10\}$.

Table 2

Average computation times for the RoA computations in the double integrator example (Fig. 2).

	N	ours (32)	Chen et al. (2024)	Langson et al. (2004)	Köhler et al. (2019)	Lorenzen et al. (2019)	Lu and Cannon (2019)	Bujarbaruah et al. (2021) Bujarbaruah et al. (2022)
Time [ms]	5	33.9	27.4	53.5	5.3	644.2	7.9	11.2 28.3
	10	99.5	83.1	107.6	6.36	1958.8	10.0	47.6 104.9

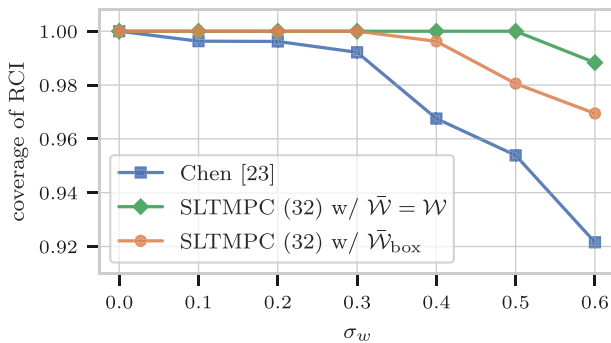


Fig. 3. RoA relative to the maximal RCI set in percent for (32) and Chen et al. (2024), where we vary σ_w and the choice of $\tilde{\mathcal{W}}$, but keep $\epsilon_A = \epsilon_B = 0.1$ and $N = 5$ fixed.

$[-7, 0]^T$. The average cost of the closed-loop trajectories are 2143.7 for Chen et al. (2024) and 1951.1 for (32), which is a 9% improvement. This behavior is expected since (32) is applied in receding horizon, while the method in Chen et al. (2024) is applied for a shrinking horizon. This is most evident in the first 5 timesteps, where Chen et al. (2024) exhibits abrupt changes in both the state and input trajectories.

6.2. Vertical take-off and landing (VTOL) vehicle

Next, we show-case the proposed filter-based SLTMPC on a VTOL vehicle model. We use Bouffard, Aswani, and Tomlin (2012) to derive the dynamics of a VTOL vehicle – inspired by the RockETH (Spannagl et al., 2021) – moving in a two-dimensional

plane. We obtain the discrete-time dynamics

$$\begin{aligned} p_{i+1}^x &= p_i^x + \Delta_t v_i^x, & p_{i+1}^z &= p_i^z + \Delta_t v_i^z, \\ v_{i+1}^x &= v_i^x + \Delta_t k_1 \theta_i, & v_{i+1}^z &= v_i^z + \Delta_t u_i^z, \\ \theta_{i+1} &= \theta_i + \Delta_t \omega_i, & \omega_{i+1} &= \Delta_t k_2 \theta_i + \omega_i + \frac{\Delta_t}{I} u_i^\theta, \end{aligned}$$

where $p^{x/z}$ and $v^{x/z}$ denote the position and velocity of the VTOL vehicle in x and z direction, respectively, θ and ω are the angle and angular velocity of the vehicle with respect to the upright position, u^z , u^θ denote the control inputs, the parameters k_1 , k_2 define the interaction between the translational and rotational dynamics, I is the inertia, and Δ_t is the discretization time. We choose all parameters such that they mimic the RockETH (Spannagl et al., 2021) and constrain its position to $[-15 \text{ m}, 15 \text{ m}]$ and $[0 \text{ m}, 15 \text{ m}]$ in the x and z direction, respectively. The control inputs are constrained to $[-5 \text{ N}, 5 \text{ N}]$ and $[-0.5 \text{ Nm}, 0.5 \text{ Nm}]$ for u^z and u^θ , respectively. The interaction parameters k_1 , k_2 are uncertain and we assume a $\pm 5\%$ modeling error on each of them, which we model as the parametric uncertainty Δ_A . Additionally, we assume wind gusts of $\pm 2 \text{ m/s}$ in x direction, which we model as the additive disturbance w . We assume no uncertainty in the input matrix B nor any uncertainty affecting the z direction. For the given uncertainties, it is not possible to compute a polytopic RPI set for dynamics (1), therefore rendering both (23) and the method in Chen et al. (2024) inapplicable. However, we can compute an RPI set for auxiliary dynamics (8) with only the additive disturbance set $\tilde{\mathcal{W}} = \{\tilde{w} \in \mathbb{R}^6 \mid \|\tilde{w}\|_\infty \leq 0.075\}$. In the following, we apply (32) and the asynchronous computation scheme (45), (42) with memory size $M = 4$ to the VTOL vehicle for a horizon of $N = 10$. We show the closed-loop trajectories in the $x - z$ plane for (32) and the primary process (45) in Fig. 5 and note that both (32) and the primary process with asynchronously computed tubes (45) are able to steer the vehicle to the origin. Due to the secondary process (42) only updating the tubes every 10th timestep, the SLTMPC with asynchronous

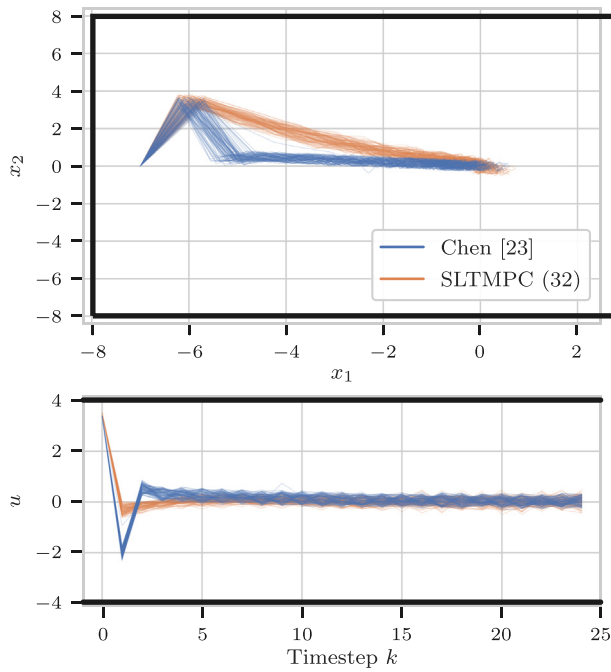


Fig. 4. Closed-loop trajectories of length 25 for Chen et al. (2024) and (32). The average costs are 2143.7 for Chen et al. (2024) and 1951.1 for (32).

computation has a higher average closed-loop cost of 5163.6 compared to 4799.4 for (32). However, primary process (45) is significantly faster than (32), with a mean computation time of 7.79 ms compared to 490.75 ms for (32). Fig. 5 also shows the convex combination parameters λ_m for the closed-loop simulations. The memory slots M_0 and M_1 are both initialized with the tubes computed by (42) for initial condition $[x, z]^T = [10, 12.5]^T$, memory slot M_3 is initialized with tubes computed by (42) for initial condition $[x, z]^T = [0, 0]^T$, and memory slot M_2 is left empty. The memory is then updated by Algorithm 2 in every timestep and by Algorithm 1 every ten timesteps, where the new tubes are stored in M_1 and the previous ones are copied to M_2 . In Fig. 5, these updates are clearly visible by the spikes in the convex combination variables. The spikes occur because we copy the previous tubes to M_2 , i.e., when the tubes in M_1 (blue lines) are updated, it is beneficial to fuse them with the previous ones in M_2 (green lines). However, after a few timesteps it is preferable to mainly use the new tubes, which is evident by increasing λ_1 (blue) and decreasing λ_2 (green) towards the next tube update. For $k \geq 50$, the trajectories are approaching the origin and the tubes in M_3 (yellow lines) are increasingly used, which is expected since they are optimized for the region around the origin. Finally, we discourage the use of the fallback tubes in M_0 (red lines) with a regularization cost on λ_0 , which is evident by their low usage, i.e., these tubes are only used if necessary to retain feasibility.

7. Conclusions

This paper proposed filter-based system level tube-MPC (SLTMPC) for constrained discrete-time linear systems with additive disturbances and model uncertainties. We introduced a new terminal controller design combined with an online optimized terminal set, which allowed us to show rigorous closed-loop guarantees in receding horizon for the first time for this type of MPC method. Additionally, we introduced a new asynchronous computation scheme for filter-based SLTMPC, which significantly reduces its computational demand by separating the tube and

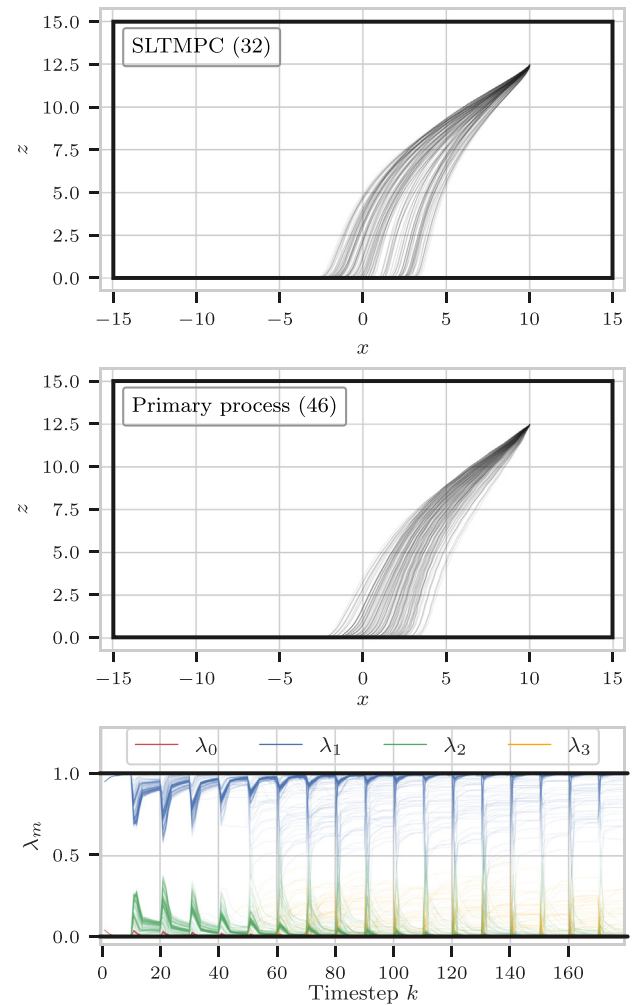


Fig. 5. Closed-loop trajectories in the x - z plane of the VTOL vehicle with SLTMPC (32) (top) and the primary process (32) (middle), and the corresponding convex combination parameters λ_m , $m = 0, \dots, M-1$ (bottom).

nominal trajectory optimizations into different processes. Finally, we showed the benefits and effectiveness of the proposed methods on two numerical examples. In future work, the proposed method can be extended to other uncertainty descriptions and potentially online-updated uncertainty sets.

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